10TH CLASS CHEMISTRY — NEW SYLLABUS 2025-26 (PECTAA)

Chapter 17 — Reaction Kinetics

Solved Quick Checks + Exercise | New Book (PECTAA 2025-26)

Quick Checks (Solved)

Quick Check 17.1

1. How will you increase the frequency of collisions?

The frequency of collisions is the number of collisions taking place per unit time, and it can be increased in the following ways.

(i) Increase the concentration of the reactants, so that more particles are present per unit volume and they meet each other more often.

(ii) Increase the temperature, because at a higher temperature the particles gain kinetic energy, move faster and collide more frequently.

(iii) For gaseous reactants, increase the pressure, which brings the particles closer together.

(iv) Increase the surface area of a solid reactant by powdering it, so that a greater number of its particles is exposed for collision.

2. What happens to the reactant particles after collision?

In most collisions the reactant particles do not have enough energy, so they simply bounce back without any change; such collisions are called ineffective collisions. Only those collisions in which the particles possess energy equal to or greater than the activation energy of the reaction lead to the breaking of old bonds and the formation of products; these are called effective or successful collisions. In such a collision the particles rise to a high-energy state and then change into the stable products.

Quick Check 17.2

1. Why is a catalyst used in a reaction preferably taken in the powdered form?

The catalytic action takes place only on the surface of the catalyst, where the reactant particles are adsorbed. A finely powdered catalyst has a much greater surface area than the same mass of a solid piece, so more reactant particles can be adsorbed on it at the same time and more successful collisions take place. In this way a small amount of a powdered catalyst gives a much faster reaction than a large piece of the same catalyst.

2. How do you increase the number of successful collisions?

The number of successful collisions can be increased by increasing the concentration of the reactants, by increasing the pressure of gaseous reactants, by increasing the temperature, and by increasing the surface area of solid reactants. Adding a suitable catalyst also increases the number of successful collisions, because it provides an alternative path of lower activation energy, so a greater fraction of the collisions becomes effective.

3. How do the concentrations of the products change during a reaction? Explain with a graph.

At the start of the reaction the concentration of the products is zero. As the reaction proceeds, the products are formed and their concentration increases rapidly at first, because the rate of the reaction is high at the beginning. As the reactants are used up, the rate of the reaction decreases, so the curve becomes less steep. When the reaction is complete, the concentration of the products reaches its maximum value and the curve becomes flat (horizontal).

On a concentration–time graph, the curve for the products therefore starts from the origin, rises steeply and then levels off; the curve for the reactants starts from a high value and falls, and both curves become flat at the end of the reaction.

Quick Check 17.3

1. How would you determine the best time of fruit harvesting?

The ripening of fruit is a set of chemical reactions which are catalysed by enzymes, for example the conversion of starch into sugar and the production of ethene gas. By studying the rate of these reactions, that is by measuring the amount of sugar, the colour, the firmness and the ethene released, food scientists can judge the stage of ripening. When these readings show that the fruit has reached its full colour, taste and sweetness, that time is chosen as the best time of harvesting, so that the fruit reaches the market in good condition.

2. Why are some fruits and vegetables destroyed early during summer than during winter?

High temperature increases the kinetic energy of the particles, so the enzyme-catalysed reactions of ripening and the growth of microorganisms become faster. In summer the warm weather speeds up these reactions, so the fruit and vegetables soften and spoil early. In winter the low temperature slows down the enzymatic reactions and the growth of microbes, acting like natural refrigeration, so the same fruit and vegetables stay fresh for a longer time.

Exercise (Solved)

A. Multiple Choice Questions

1. The number of collisions per unit volume of the reaction mixture is called:

(a) Collision energy
(b) Activation energy
(c) Collision frequency
(d) Collision force

Answer: (c) — Collision frequency is the number of collisions taking place per unit volume in unit time, and the rate of reaction depends upon it.

2. If the reactants possess an energy higher than the activation energy, then the reaction will be:

(a) Slow (b) Fast

(c) Not affected (d) Instantaneous

Answer: (b) — Particles having energy higher than the activation energy cross the energy barrier easily, so more effective collisions take place and the reaction is fast.

3. Which of the following explains the increase in rate of reaction in the presence of a catalyst?

(a) Catalyst provides extra energy to the reactant molecules

(b) Catalyst provides an alternative pathway which lowers the activation energy.

(c) Catalyst increases the collision frequency

(d) Catalyst decreases the collision frequency

Answer: (b) — A catalyst does not supply energy; it provides an alternative path of lower activation energy, so more particles can cross the barrier.

4. Which of the following statements is correct?

(a) Collisions with energy equal to or greater than the activation energy lead to reaction.

(b) Collision frequency is not related to the reaction rate.

(c) All collisions lead to a reaction.

(d) Collisions with energy less than the activation energy lead to the reaction.

Answer: (a) — This is the basic condition of the collision theory; only such collisions are effective.

5. When a reaction proceeds ahead, how do the concentrations of reactants and products change?

(a) Concentration of reactants increases and that of the products decreases.

(b) Concentration of reactants decreases and that of the products increases.

(c) Concentration of both reactants and products decreases.

(d) Concentration of both reactants and products increases.

Answer: (b) — Reactants are used up during the reaction while products keep forming, so the concentration of reactants decreases and that of the products increases.

B. Short Answer Questions

17.1 What is a successful collision?

A collision in which the colliding particles possess energy equal to or greater than the activation energy of the reaction is called a successful or effective collision. In such a collision the old bonds of the reactants break and new bonds are formed, so products are obtained. Most of the collisions are ineffective, because the particles possess less than the activation energy and simply bounce back unchanged.

17.2 How does increase in the temperature increase the rate of a reaction?

When the temperature is increased, the kinetic energy of the reacting particles increases, so they move faster and collide more often and with greater force. A greater fraction of the particles now possesses energy equal to or greater than the activation energy, so the number of successful collisions per second increases and the reaction becomes faster. For many reactions the rate becomes almost double with every 10 °C rise in temperature.

17.3 Define activation energy.

The minimum energy which the colliding particles must possess to start a chemical reaction is called the activation energy of that reaction. It is the energy needed to break the bonds present in the reactant molecules. Particles having energy equal to or greater than this value react, while particles having less energy simply bounce back without reaction.

17.4 Why does the burning of sulphur proceed slower in air than in pure oxygen?

The rate of a reaction depends upon the concentration of the reactants, that is, on the number of particles per unit volume. Air contains only about 21 % oxygen, so the concentration of oxygen in air is low and only a small number of oxygen molecules is available to collide with the sulphur. In pure oxygen the concentration of oxygen is much higher, so the number of effective collisions increases greatly and sulphur burns much faster.

17.5 Why a catalyst used in a reaction is preferably taken in the powdered form?

The action of a catalyst takes place on its surface, where the reactant molecules are adsorbed and their bonds are weakened. When the catalyst is taken in powdered form, its surface area becomes very large, so a greater number of reactant particles can be adsorbed on it at the same time. This increases the number of successful collisions and the reaction becomes faster even with a small amount of the catalyst.

17.6 Why is the rate of a reaction often very fast at the beginning of the reaction?

At the beginning the concentration of the reactants is at its maximum, so the number of particles per unit volume is very large and the particles collide very frequently. The number of successful collisions per second is therefore the greatest, and the reaction is very fast. As the reaction proceeds, the reactants are used up, their concentration falls, the number of effective collisions decreases and the rate of the reaction becomes slower and slower until it finally stops.

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17.7 Magnesium does not react with air at room temperature but it reacts very fast at high temperature giving intense white light. Explain.

At room temperature the particles of magnesium and oxygen possess only average kinetic energy, which is less than the activation energy of the reaction, so their collisions are ineffective and no reaction takes place. When magnesium is strongly heated, its particles gain a large amount of kinetic energy, which becomes greater than the activation energy. Now the collisions become successful, the reaction starts and the heat released by the reaction itself keeps it going, so magnesium burns rapidly with intense white light forming white magnesium oxide.

$$2Mg(s) + O_{2(g)} \rightarrow 2MgO(s)$$

17.8 What happens to the reactants after they climb the energy hill during a reaction?

When the reactant particles possess activation energy, they reach a high-energy unstable state at the top of the energy hill. In this state the old bonds are broken and new bonds begin to form. The particles then slide down the other side of the hill and change into the stable products, and their potential energy falls to a lower value. If the products have lower energy than the reactants, the extra energy is released as heat, so the reaction is exothermic.

17.9 How does a catalyst lower the activation energy of a reaction?

A catalyst does not supply energy to the reactants; instead it provides an alternative pathway for the reaction which has a lower activation energy. The reactant molecules are adsorbed on the surface of the catalyst and their bonds are weakened. In this way less energy is needed to break the bonds, so a much greater fraction of the collisions becomes successful at the same temperature and the rate of the reaction increases. The catalyst itself is recovered unchanged at the end of the reaction.

17.10 Give any two features of the catalytic action of an enzyme.

(i) Specificity: each enzyme has active sites of a definite shape, so a particular enzyme can catalyse only one specific reaction. The reactant molecule fits into the active site like a key in a lock.

(ii) Reusability: after the products are formed they leave the enzyme, and the enzyme becomes free to bind another reactant molecule. Therefore an enzyme is not used up or consumed in the reaction it catalyses.

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C. Constructed Response Questions

17.1 In what different ways you can increase the successful collisions between the particles of the reactants?

Successful collisions can be increased by raising the concentration of the reactants, which increases the number of particles per unit volume. The pressure of gaseous reactants can be increased so that their particles come closer. Solids can be ground into powder, because the reaction takes place only on the surface where the particles of the reactants meet. The temperature can be raised, because at high temperature the particles move faster and a greater fraction of them possesses the activation energy. Finally, a suitable catalyst can be added, which provides a path of lower activation energy so that more collisions become effective.

17.2 Give an example of a reaction which proceeds with the gain in mass.

When a ribbon of magnesium is burnt in an open crucible in air, the white powder of magnesium oxide formed has a greater mass than the original magnesium ribbon. This is because magnesium combines with the oxygen of the air, which is added to the product.

$$2Mg(s) + O_{2(g)} \rightarrow 2MgO(s)$$

The reaction appears to gain mass only because a gas from outside has been absorbed. If the same reaction is carried out in a closed container, the total mass before and after the reaction remains exactly the same, in accordance with the law of conservation of mass. On the other hand, when marble chips react with dilute hydrochloric acid in an open container, the mass appears to decrease because carbon dioxide gas escapes from the mixture.

17.3 From where do molecules get energy to attain higher energy state?

The molecules get this energy from their own kinetic energy and from the heat of the surroundings. All particles are in continuous motion and therefore possess kinetic energy; when two particles collide, a part of this kinetic energy is converted into potential energy, so the particles reach the high-energy state at the top of the energy barrier. Heating the reaction mixture increases the kinetic energy of the particles, so a greater number of particles is able to attain the activation energy and cross the barrier.

17.4 How does the presence of V_2O_5 catalyst lower the activation energy of the following reaction? $2SO_2 + O_2 \rightarrow 2SO_3$

Vanadium pentoxide provides a surface on which the molecules of sulphur dioxide and oxygen are adsorbed. On this surface their bonds are weakened and they react through an alternative pathway which needs much less energy, so the activation energy of the reaction is lowered. The molecules of SO_3 formed then leave the surface of the catalyst, and the catalyst is left unchanged to adsorb the next molecules. This is the contact process used for the manufacture of sulphuric acid.

17.6 Explain the catalytic action of an enzyme.

Enzymes are biological catalysts; they are proteins which speed up specific reactions in living bodies. Each enzyme has active sites of a particular shape, and the reactant molecule binds to these sites like a key fitting into a lock. After binding, the particular bonds in the reactant molecule are weakened, so the reaction takes place quickly at ordinary body temperature with a much lower activation energy. The products then leave the enzyme, and the free enzyme is ready to bind another molecule of the reactant. Because of the specific shape of the active sites, one enzyme catalyses only one particular reaction.

17.7 If you desire to stop a reaction going on at 60 °C, what action you will take?

The reaction should be cooled immediately, for example by placing the vessel in ice water. By lowering the temperature the kinetic energy of the reacting particles is decreased, so the number of particles possessing the activation energy becomes very small and the effective collisions stop. In this way the reaction can be stopped. The reaction could also be stopped by removing one of the reactants or by removing the catalyst, if one is being used.

D. Descriptive Questions

17.1 Explain the effect of surface area on the rate of a reaction.

A reaction involving a solid takes place only on the surface of the solid, where its particles come in contact with the particles of the other reactant. The larger the surface area of the solid, the greater is the number of its particles exposed for collision, so the number of effective collisions per second increases and the rate of the reaction becomes faster. That is why finely divided solids react much more rapidly than large pieces of the same substance.

For example, aluminium foil reacts slowly with warm sodium hydroxide solution, but aluminium powder reacts rapidly even with cold sodium hydroxide solution and gives hydrogen gas. Similarly, a large piece of wood burns slowly, while small pieces or wood shavings catch fire and burn rapidly. In the laboratory, marble chips react slowly with hydrochloric acid, but powdered marble reacts very quickly.

17.2 Describe the main points of the collision theory of reaction rate.

(i) A chemical reaction takes place only when the particles of the reactants collide with one another. However, only a very small fraction of the collisions leads to the formation of products; these are called effective or successful collisions.

(ii) The rate of a reaction depends upon the number of effective collisions per unit time. This number depends upon the number of particles per unit volume, the frequency of collisions, and the kinetic energy of the particles.

(iii) Only those colliding particles react which possess energy equal to or greater than the activation energy of the reaction; particles with less energy simply bounce back.

(iv) After a successful collision the particles reach a high-energy state and then change into the stable products, releasing the extra energy as heat if the products are of lower energy.

(v) Any factor which increases the number of effective collisions per second, such as concentration, pressure, surface area, temperature or a catalyst, increases the rate of the reaction.

17.3 Discuss the following factors affecting the rate of a reaction: (a) effect of concentrations of the reactants (b) effect of temperature at which a reaction is carried out.

(a) Effect of concentration: the rate of a reaction depends upon the number of particles of the reactants per unit volume. When the concentration of a reactant is increased, more particles are present in the same volume, so they collide more frequently and the number of effective collisions increases, which increases the rate. That is why coal burns slowly in air, which contains 21 % oxygen, but burns very fast in pure oxygen. In the experiment with limestone and hydrochloric acid, 2 M HCl gives carbon dioxide much faster than 1 M HCl, and the curve of volume of gas against time is steeper at the start. The rate is highest at the beginning and decreases as the reactant is consumed.

(b) Effect of temperature: when the temperature is raised, the average kinetic energy of the reacting particles increases, so they move faster and collide more often and with greater force. A much greater fraction of the particles now possesses the activation energy, so the number of successful collisions increases and the rate of the reaction rises; for many reactions the rate becomes almost double for every 10 °C rise. Food cooks faster at high temperature than at low, and iron rusts very slowly at room temperature but oxidises very fast when strongly heated.

17.4 How does chemical kinetics play an important role in the food industry?

The study of the rates of reactions and the factors which affect them is very important in the food industry. The rates of the reactions of ripening and spoilage are studied for different fruits and vegetables, and this information is used to minimise the losses caused by spoilage. By knowing these rates, food scientists can decide the best conditions of harvesting, storage and transportation of food.

Many enzymes are naturally present in food, and they cause changes in texture, flavour and nutritional value; if a fruit is known to spoil rapidly at high temperature, refrigeration is used during transport and storage to slow down these enzymatic reactions. In the same way, a knowledge of the kinetics of the oxidation of fats helps to prevent the spoilage of butter, and the rates of the reactions of microorganisms help in deciding how long a food product can be stored safely.

17.5 Write briefly about the role enzymes play in our body as biological catalysts.

Enzymes are biological catalysts which are basically proteins, and they speed up the specific chemical reactions taking place in our body at normal body temperature. Each enzyme has active sites of a definite shape, and the reactant molecule binds to these sites; after binding, the particular bonds of the reactant are weakened and the reaction takes place quickly. The product formed then separates from the enzyme, which becomes free to bind another reactant molecule.

Because the active sites are highly specific, one enzyme catalyses only one particular reaction. In this way enzymes digest food, build new cells and carry out all the metabolic reactions of the body without being consumed themselves, so a very small amount of an enzyme can catalyse a reaction for a long time.

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